NEUROQUEST

Adaptive EEG-Based Brain-Computer Interface Gaming for Cognitive Enhancement in Neurocognitive Disorders

INTRODUCTION

Neurocognitive diseases and disorders like ADHD severely impact an individual's ability to focus. ADHD is highly prevalent, affecting approximately **7.6%** of children and adolescents and **6.76%** of adults globally. Beyond internal struggles, these conditions lead to higher rates of unemployment and social difficulties. While stimulants are effective, they often cause side effects like insomnia and cardiovascular risks. Behavioral therapy is an alternative but is often hindered by high costs and intensive time commitments.

BACKGROUND

Existing "brain training" digital interventions, such as **EndeavorRX** or **Lumosity**, often face criticism for the "proxy problem". This means users may improve at specific in-game motor tasks without the training transferring to underlying neural attention deficits or real-world behavioral symptoms. Clinical neurofeedback uses direct brain signals but remains largely inaccessible to the general public due to lack of insurance coverage and costs ranging from **$100-$200 per session**.

Figure 1. In-Game Screenshot of EndeavorRX

PURPOSE

The objective of NeuroQuest is to create an accessible, BCI-based gaming system that directly trains the brain's attentional networks. By "closing the loop" between a user's measurable brain state and game feedback, the system addresses the proxy problem and avoids the need for motor-skill-based inputs. Utilizing open-source hardware allows for a scalable, community-driven approach to cognitive therapy.

METHODOLOGY

Stage 1: Hardware Integration

1. An OpenBCI 16-channel Ultracortex is utilized for signal acquisition.
2. A Cyton + Daisy Board transmits data from electrodes to the laptop via a USB dongle.
3. Focus is primarily detected via the O1, O2, P3, and P4 nodes located near the Occipital lobe.

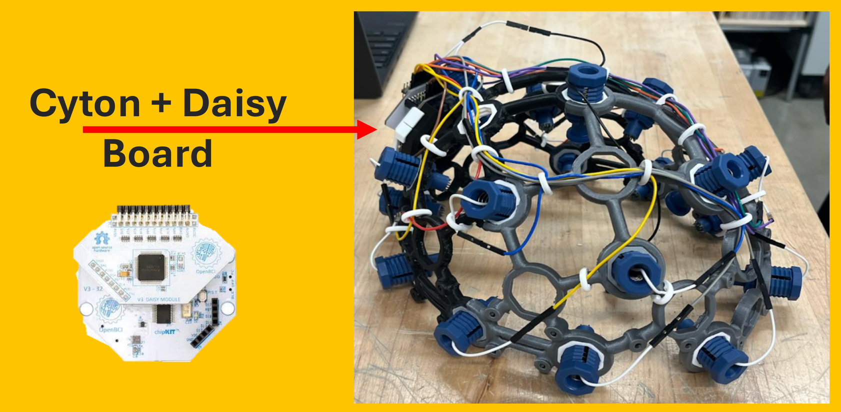

Figure 2. OpenBCI UltraCortex (16 Channel) w/ attached Cyton + Daisy Board

Stage 2: Software & Preprocessing

- The system uses Python 3.11 for backend models and Unity 6 (C#) for the game environment.
- BrainFlow handles data streaming, MNE manages signal processing, and TensorFlow powers the machine learning models.
- Signals are sampled at 250 Hz and passed through a 0.5-45 Hz bypass filter.

Stage 3: Machine Learning Models

1. **EEG-to-Game Input Model:** Uses Steady-State Visually Evoked Potential (SSVEP). By flickering stimuli at specific frequencies (e.g., 8 Hz), the model uses Fast Fourier Transforms to translate brain rhythms into game commands.
2. **Focus-Adaptation Model:** Employs Higuchi Fractal Dimensions (HFDs) to measure signal complexity. Higher FD values indicate higher concentration.

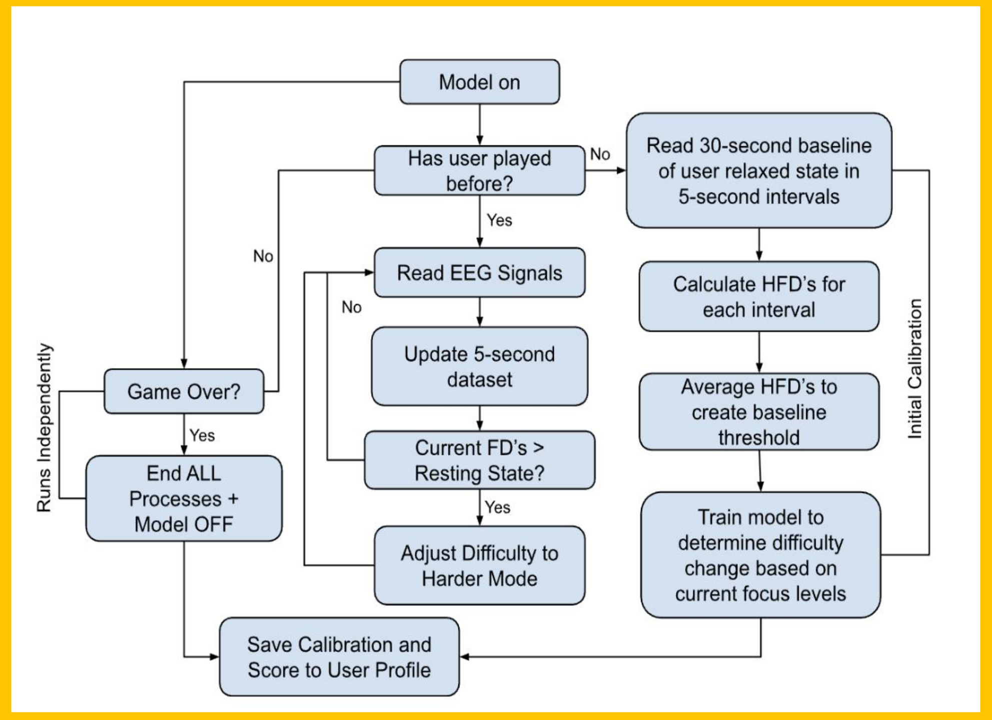

Figure 3. Flowchart of Focus Adaptation Model

Stage 4: FocusFactory Prototype

A 3D sorting game where users direct objects toward specific platforms. The game adjusts difficulty thresholds in real-time based on the user's FD calculated focus levels.

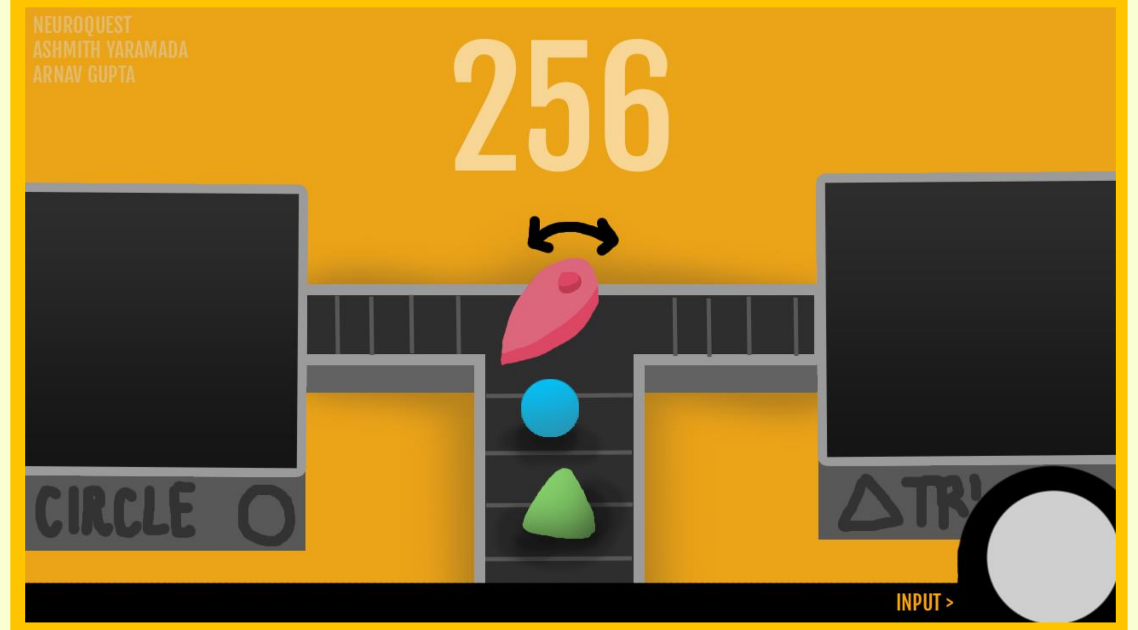

Figure 4. Initial Mockup of Focus Factory

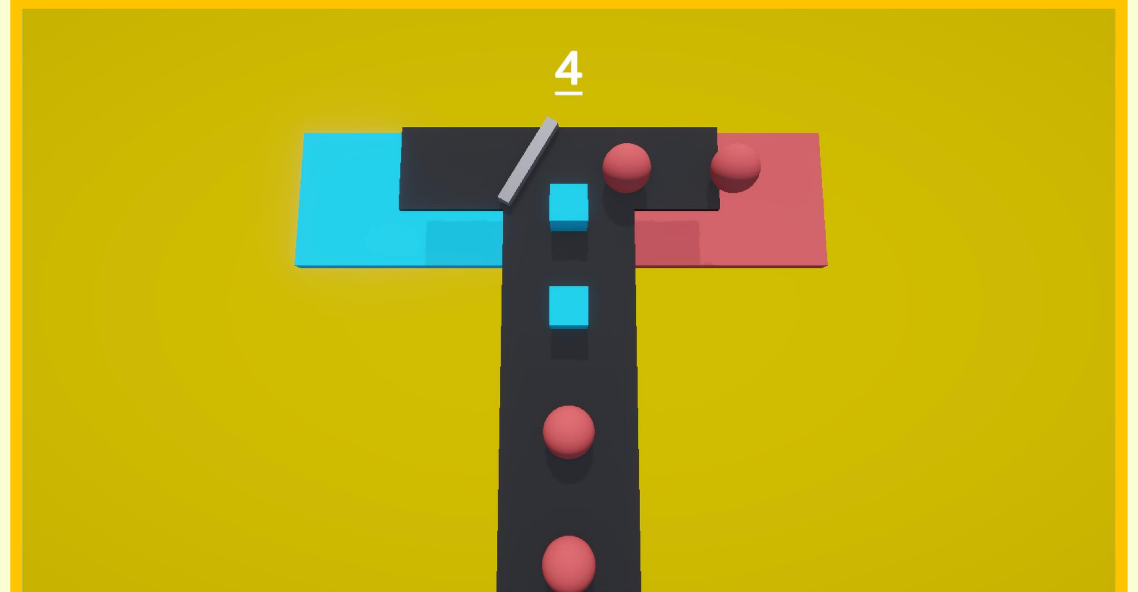

Figure 5. Functional Heuristic Focus Factory Prototype (Built in Unity 6)

RESULTS

Training Strategy	Scope	Key Preprocessing	Val. Accuracy
Baseline Prototype	Single Subject (S1)	Raw PSD (Absolute Power)	**78.10%**
Multi-Subject (Raw)	Multiple Subjects (~15)	Raw PSD (Absolute Power)	**43.00%**
Multi-Subject (Normalized)	Multiple Subjects (~15)	Relative Power (% of Total)	**75.00%**

Table 1. Input Model Development Progression

Trial	Condition	Inconclusive	Correct Prediction
Trial 1	Concentrated	20%	**80%**
Trial 2	Concentrated	20%	**75%**
Trial 3	Concentrated	30%	**70%**

Table 2. Focus Adaptation Model Performance

Normalizing absolute signal power significantly increased the model's generalizability across multiple subjects from 43% to 75%. The HFD metric currently struggles with a high margin of "inconclusive" results when the user is not focused.

CONCLUSION

NeuroQuest demonstrates that direct BCI-controlled gameplay is a viable method for training attentional networks. While the system accurately detects concentrated states, future work is needed to refine the thresholds for "unfocused" states. Future enhancements will include:

- Establishing more consistent "resting state" calibration methods.
- Implementing dual thresholds to distinguish between "active but not concentrated" states.
- Finalizing the programmatic linkage between the Python models and the live Unity engine.

RESOURCES

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